

Introduction

Sorting algorithms play an important role in data organization and optimization.

While we have previously explored foundational sorting algorithms like merge, bubble, insertion, and counting sorts, they may fall short when the size of the data is large.

Therefore, this chapter will introduce you to more advanced and specialized sorting techniques:

- **Bucket Sort**
- **Radix Sort**
- **Counting Sort**

Bucket Sort Algorithm

Bucket Sort is a sorting algorithm that divides the unsorted array elements into several groups called buckets. Each bucket is then sorted by using any of the suitable [sorting algorithms](#) or recursively applying the same bucket algorithm.

Finally, the sorted buckets are combined to form a final sorted array.

Working of Bucket Sort

Suppose we need to sort a vector of elements.

10	8	2	1	12	7	6	14	11	4
0	1	2	3	4	5	6	7	8	9

Figure: A Vector to Sort

By definition, bucket sort

1. Divides the elements into several buckets.
2. Sorts each bucket.
3. Gathers all the buckets together.

This is called the **scatter-gather approach**.

Scatter Gather Approach

The process of bucket sort can be understood as a **scatter-gather approach**. Here, elements are first scattered into buckets then the elements in each bucket are sorted. Finally, the elements are gathered in order.



Working of Bucket Sort

Quiz: Given an array of elements ranging from 1 to 20, what should be the approximate range of each bucket?

A: 0-6, 6-12, 12-20

B: 0-5, 5-15, 15-20

C: 1-10, 10-20, 20-30

D: 1-20, 20-40, 40-60

Quiz: Given an array of elements ranging from 1 to 20, what should be the approximate range of each bucket?

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B: 0-5, 5-15, 15-20

C: 1-10, 10-20, 20-30

D: 1-20, 20-40, 40-60



Correct Answer!

Since we are dividing the elements into three approximately equal ranges, these would be the suitable ranges for the buckets.

Was this helpful?

Yes

No

Quiz: In the bucket sort process, how are elements sorted inside each bucket?

A: Using any suitable sorting algorithm.

B: Using insertion sort only.

C: Using bubble sort only.

D: By placing the elements in the correct order initially.

Quiz: In the bucket sort process, how are elements sorted inside each bucket?

A: Using any suitable sorting algorithm.



Correct Answer!

The bucket elements can be sorted using any suitable sorting algorithm.

B: Using insertion sort only.

C: Using bubble sort only.

D: By placing the elements in the correct order initially.

Was this helpful?

Yes

No

Working of Bucket Sort

Let's summarize the concepts we have learned so far with the help of an image.



Figure: Working of Bucket Sort

Quiz: What is the basic operation of the bucket sort algorithm?

A: Dividing the elements into buckets and sorting each bucket independently.

B: Sorting the entire vector without any division.

C: Dividing the elements into buckets without sorting.

D: Selecting the smallest element and placing it at the start of the vector.

Quiz: What is the basic operation of the bucket sort algorithm?

A: Dividing the elements into buckets and sorting each bucket independently.



Correct Answer!

Bucket sort uses the scatter-gather approach, which involves dividing elements into buckets and sorting each bucket independently.

B: Sorting the entire vector without any division.

C: Dividing the elements into buckets without sorting.

D: Selecting the smallest element and placing it at the start of the vector.

Was this helpful?

Yes



No



Best Use Case for Bucket Sort

When to use bucket sort?

- The input data is uniformly distributed over a range, as each bucket will have a uniform load.
- The range of input data is known, which helps in determining the number and size of the buckets.

When not to use bucket sort?

The efficiency of bucket sort can be compromised if elements are not uniformly distributed.

Suppose you want to sort numbers between **1** and **100** using **10** buckets. If most numbers are between **1** and **10**, the first bucket will overflow while a few others might remain empty.

This imbalance can slow down the sorting process, as the overloaded bucket takes longer to sort.

Design and Advance Analysis of Algorithms

Sorting in Linear Time **Bucket Sort**



Bucket Sort

- Why counting sort is not works well for *dispersed data*?
- The idea of bucket sort is to divide the interval $[0, 1)$ into *n equal-sized subintervals*, or buckets, and then distribute the n input numbers into the buckets
- It assumes that the input is generated by a *random process* that distributes elements uniformly over the interval $[0, 1)$
- Since the inputs are *uniformly distributed* over $[0, 1)$, we don't expect many numbers to fall into each bucket
- Use Linked list as buckets to store elements

Bucket Sort

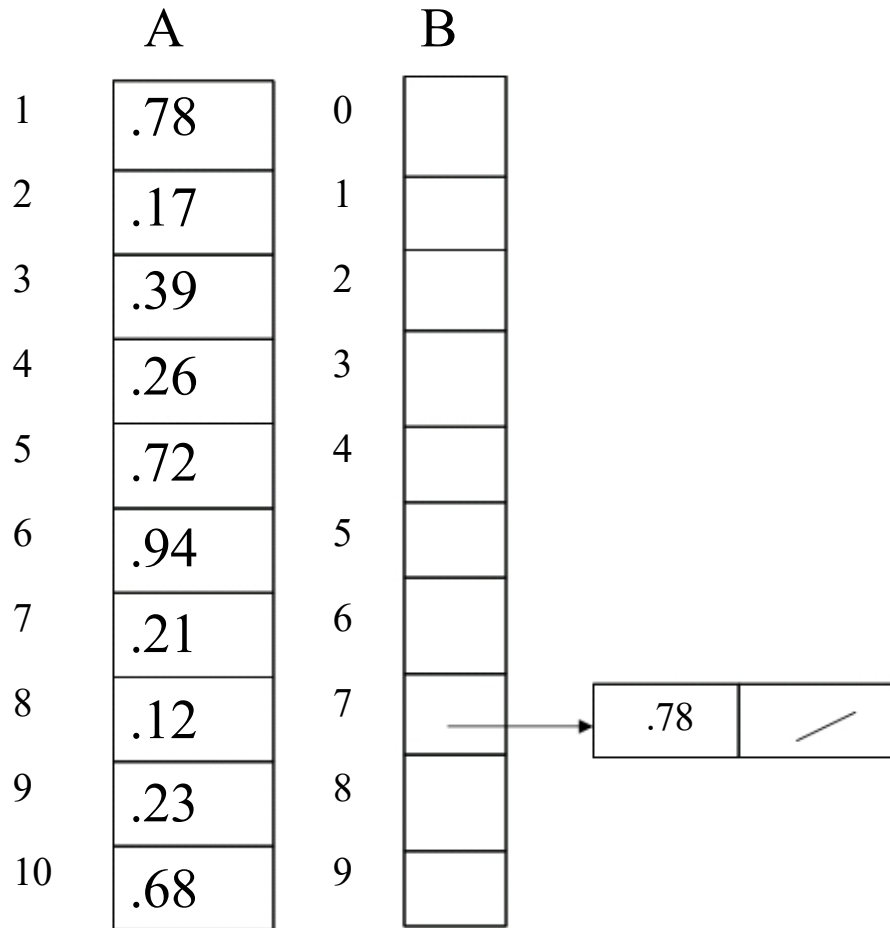
Bucket-Sort(A)

1. $n = \text{length}[A]$
2. For $i = 1$ to n
3. Do insert $A[i]$ into list $B[\lfloor n A[i] \rfloor]$
4. For $i = 0$ to $n - 1$
5. Do sort list $B[i]$ with insertion sort
6. Concatenate the lists $B[0], B[1], \dots, B[n-1]$ together in order

1 $n = 10$

2 for $i = 1$

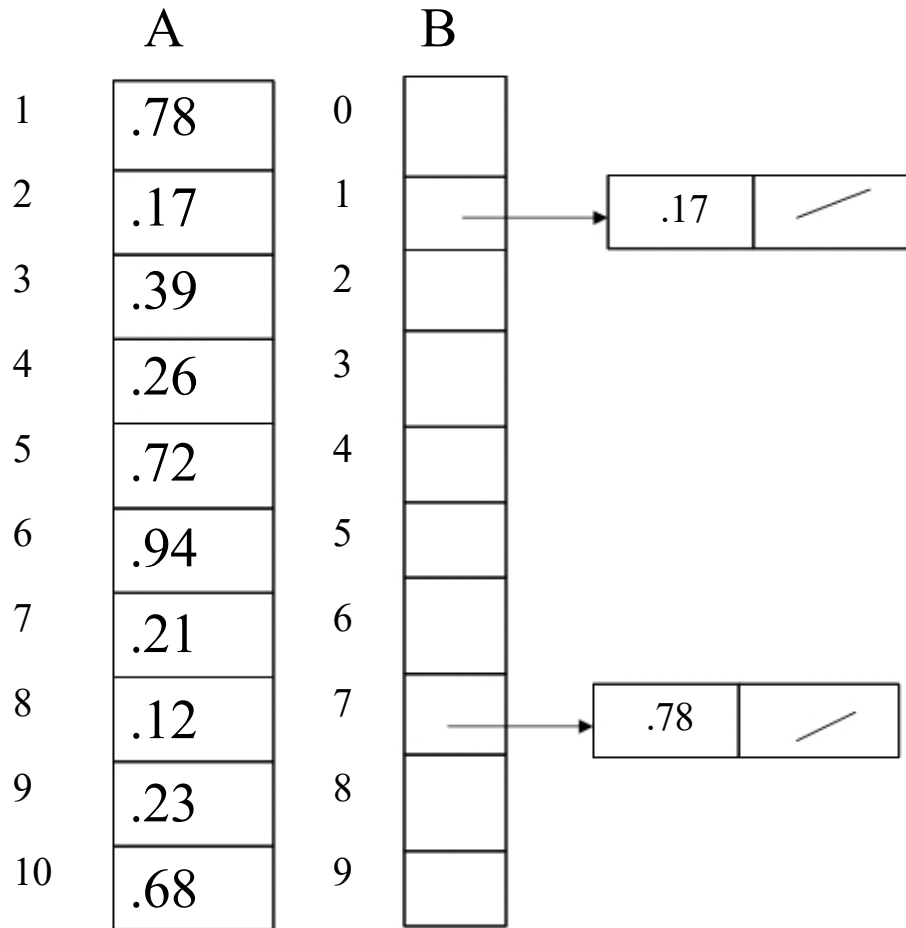
3 do insert $A[1]$ into list $B[10$
 $A[1])$



1 $n = 10$

2 for $i = 2$

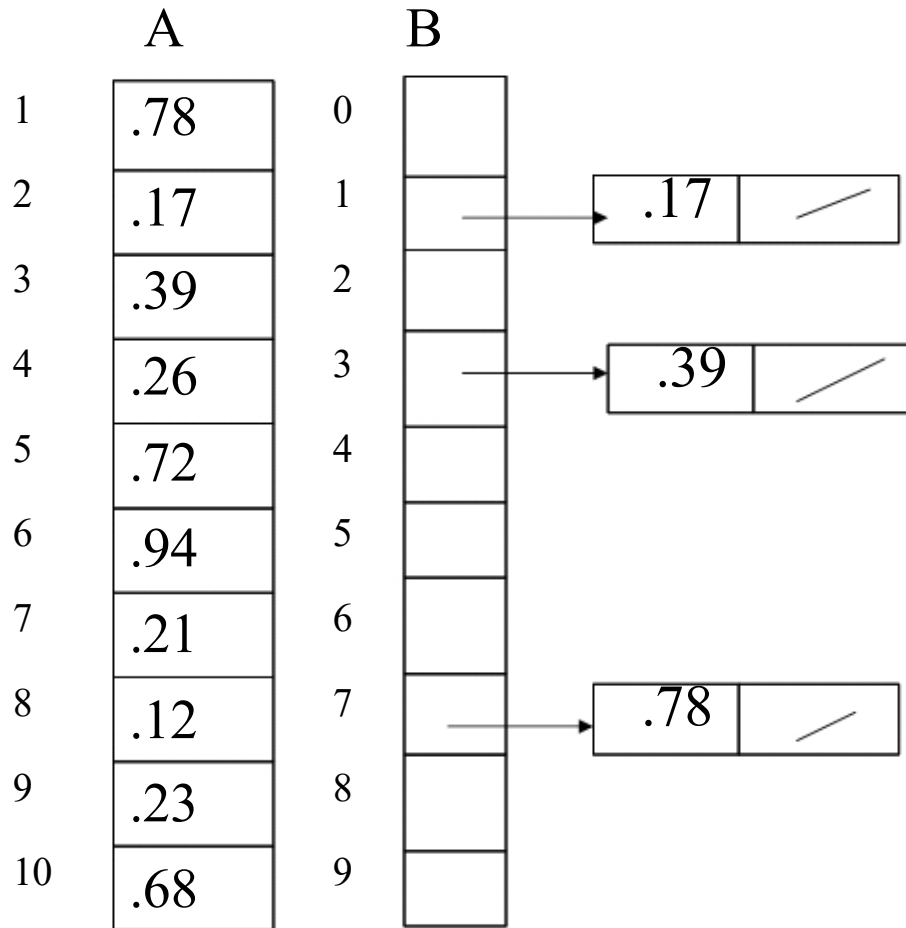
3 do insert $A[2]$ into list $B[10$
 $A[2])$



1 $n = 10$

2 for $i = 3$

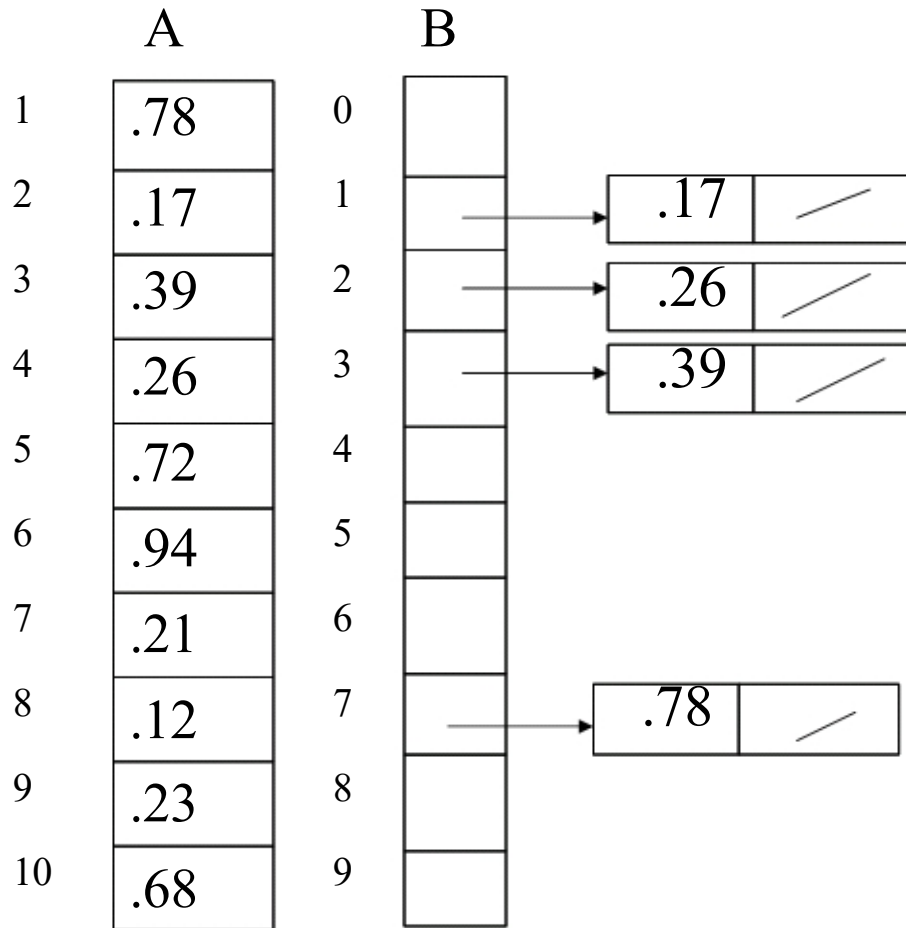
3 do insert $A[3]$ into list $B[10$
 $A[3])$



1 $n = 10$

2 for $i = 4$

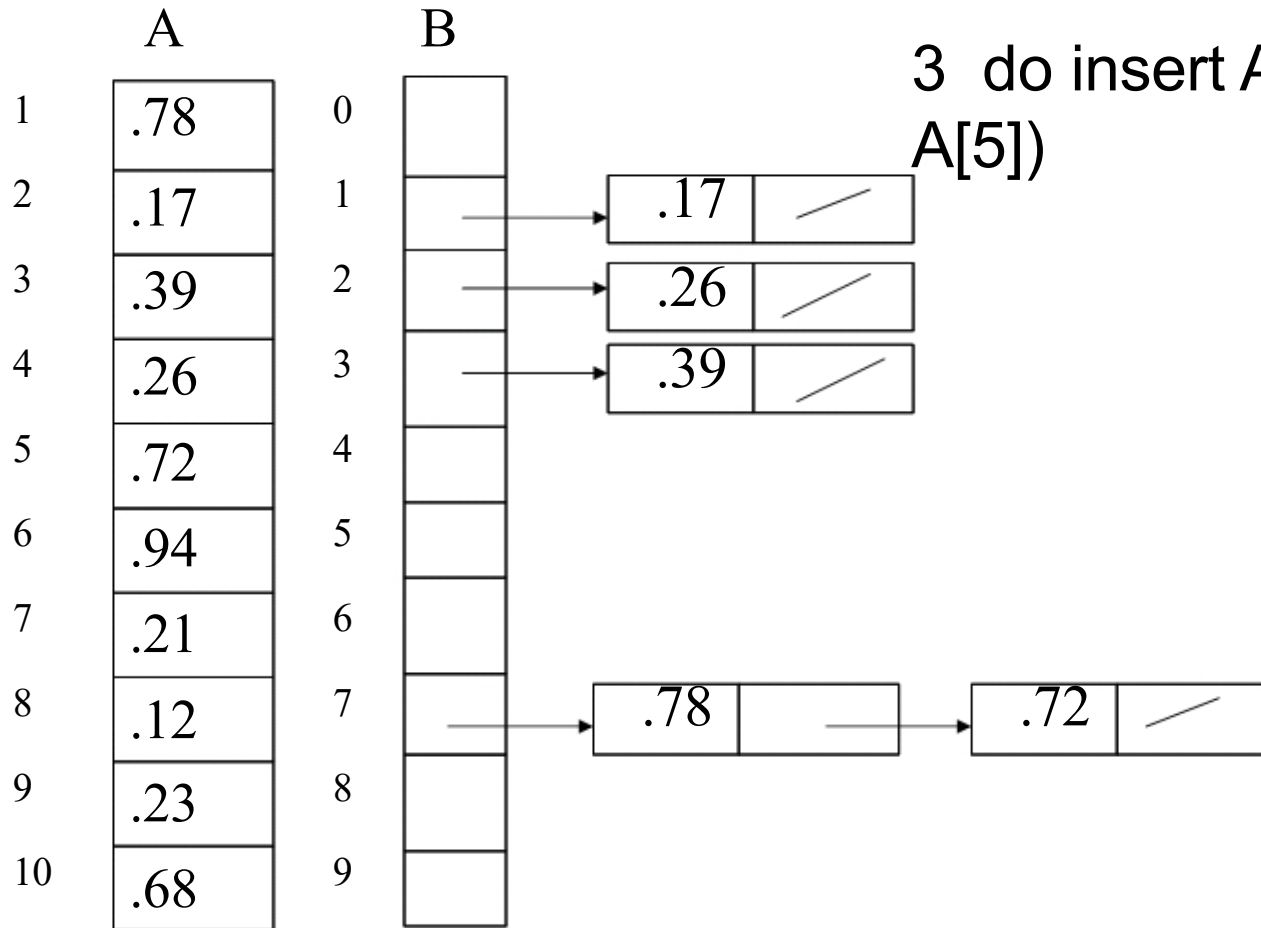
3 do insert $A[4]$ into list $B[10$
 $A[4])$



1 $n = 10$

2 for $i = 5$

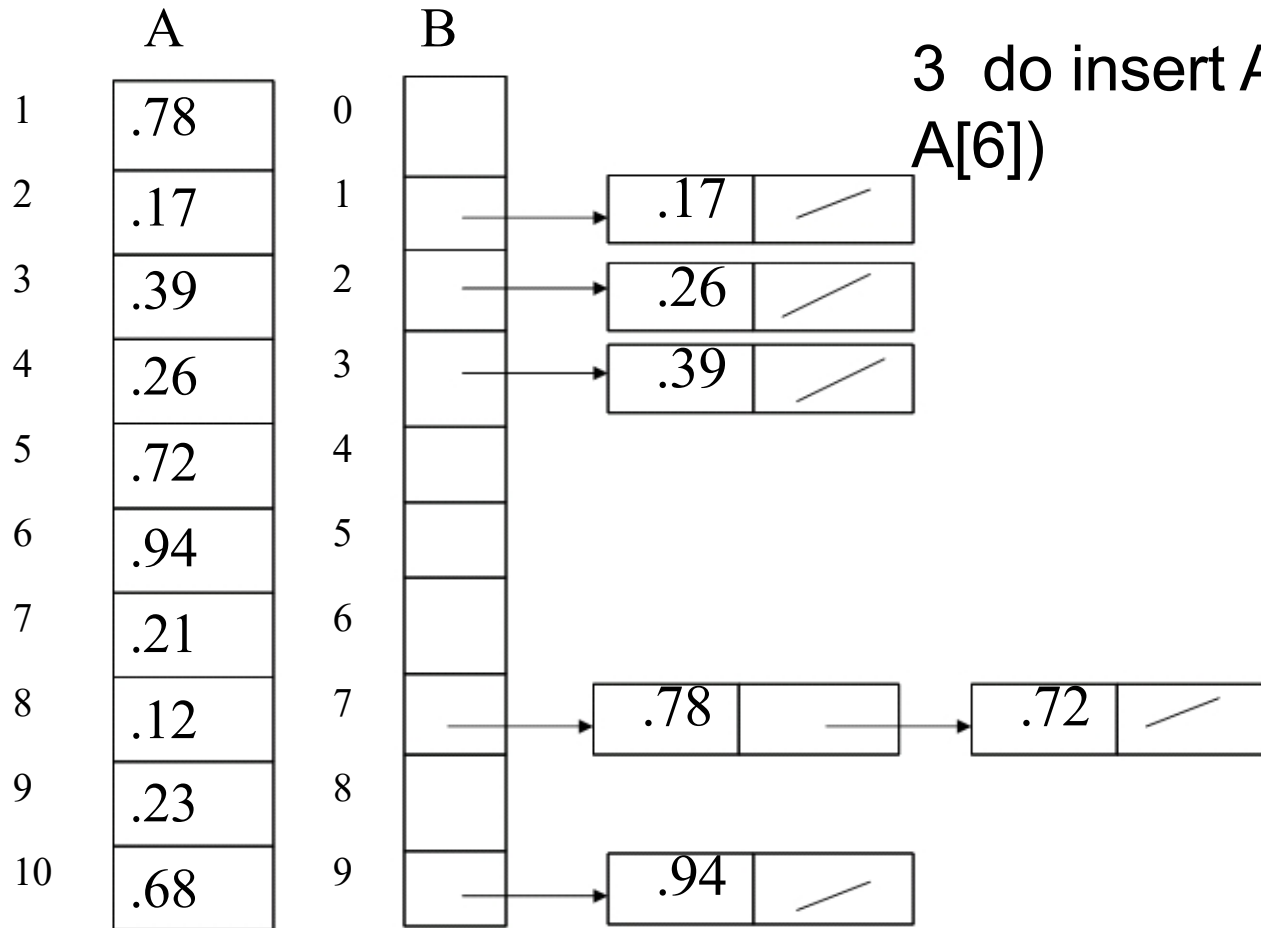
3 do insert $A[5]$ into list $B[10$
 $A[5])$



1 $n = 10$

2 for $i = 6$

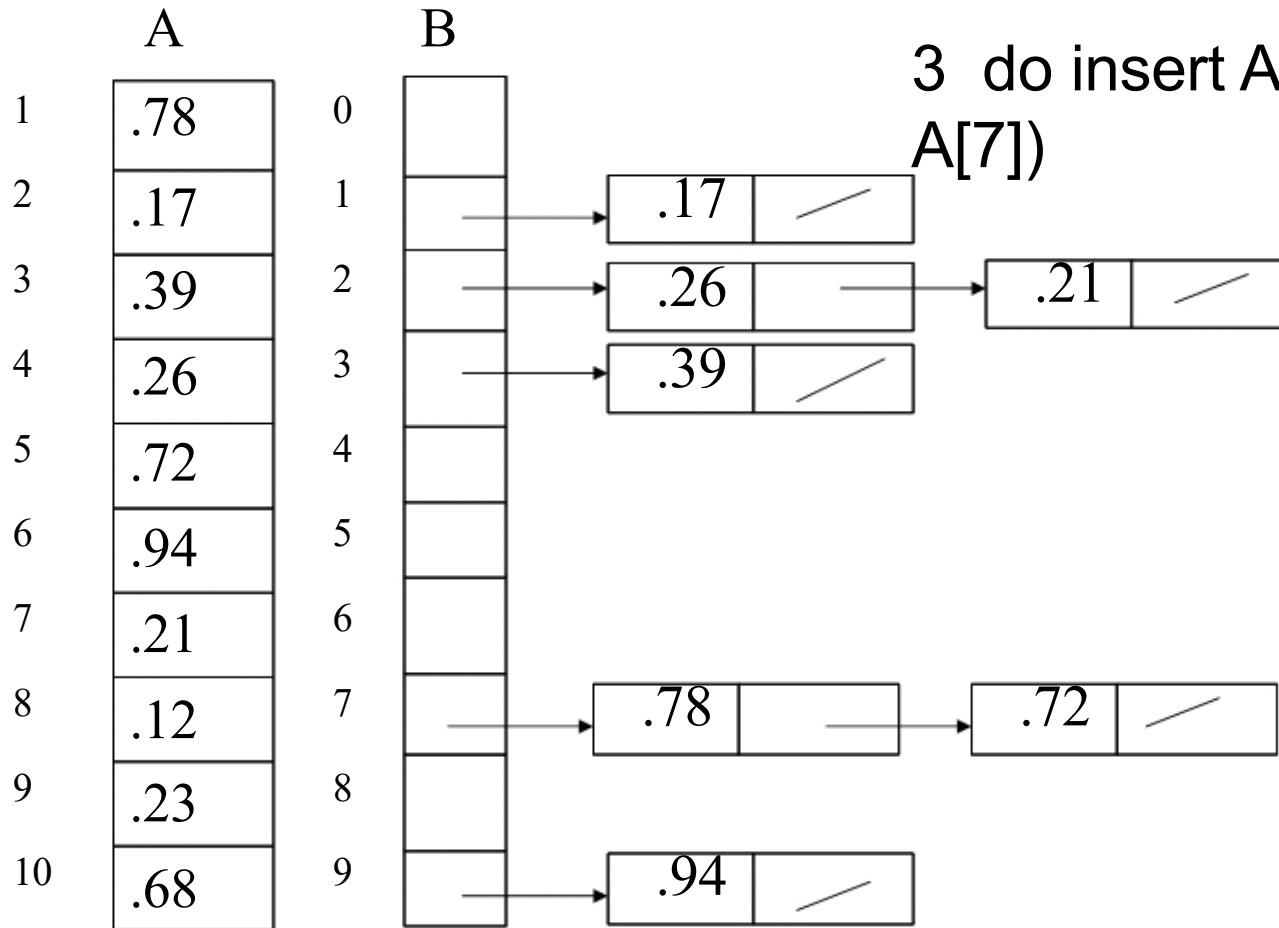
3 do insert $A[6]$ into list $B[10$
 $A[6])$



1 $n = 10$

2 for $i = 7$

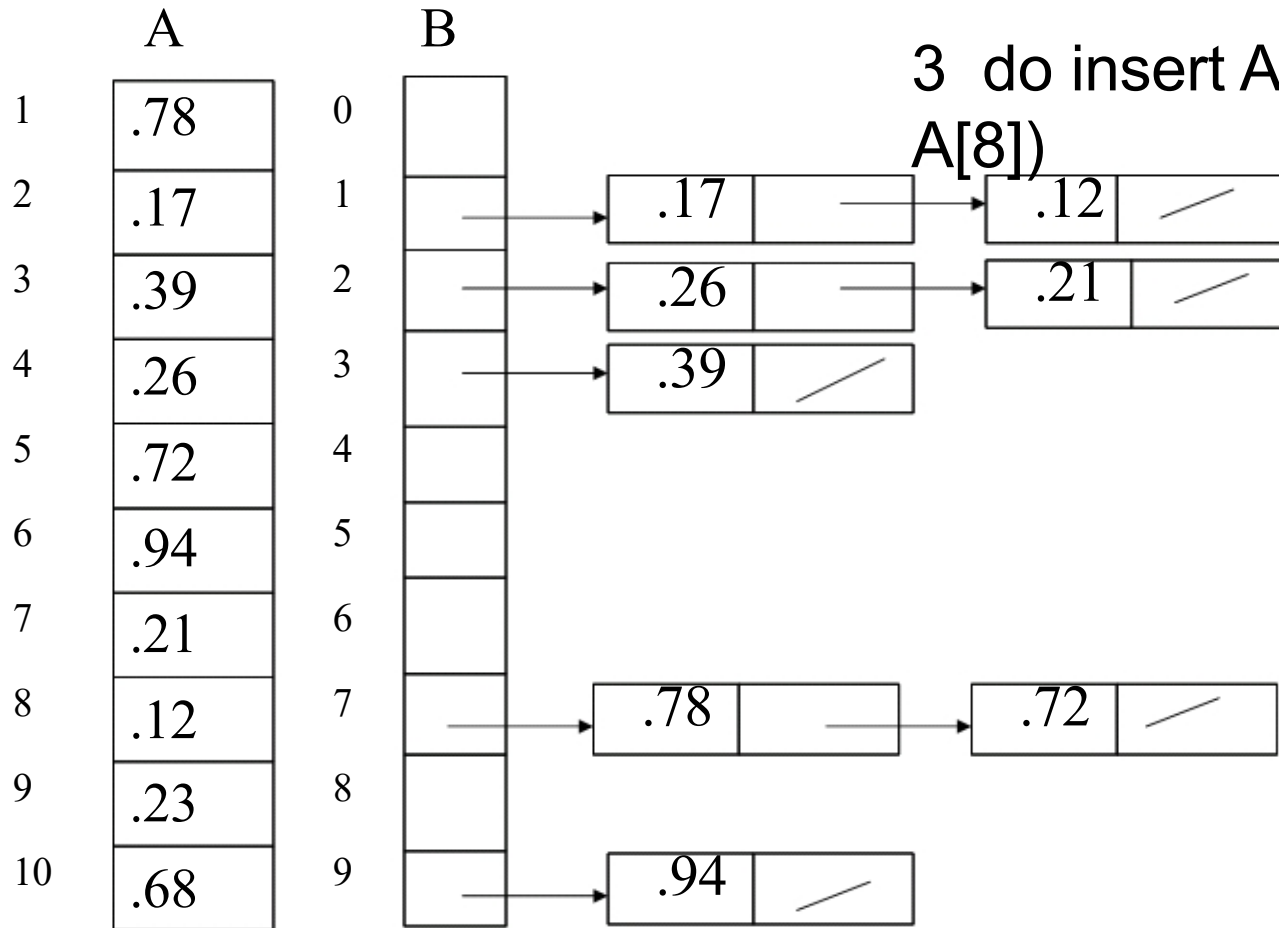
3 do insert $A[7]$ into list $B[10$
 $A[7])$



1 $n = 10$

2 for $i = 8$

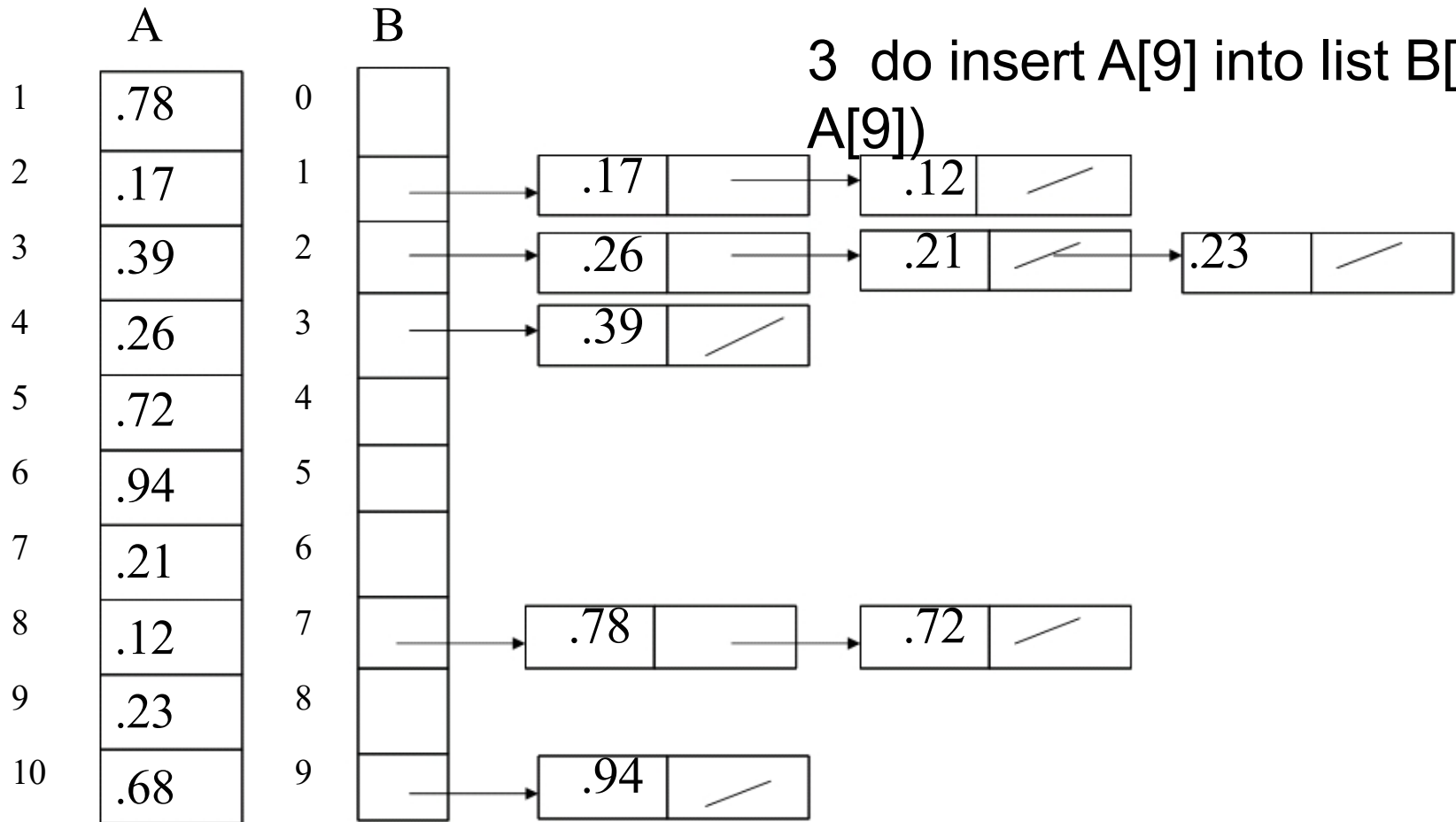
3 do insert $A[8]$ into list $B[10$
 $A[8])$



1 $n = 10$

2 for $i = 9$

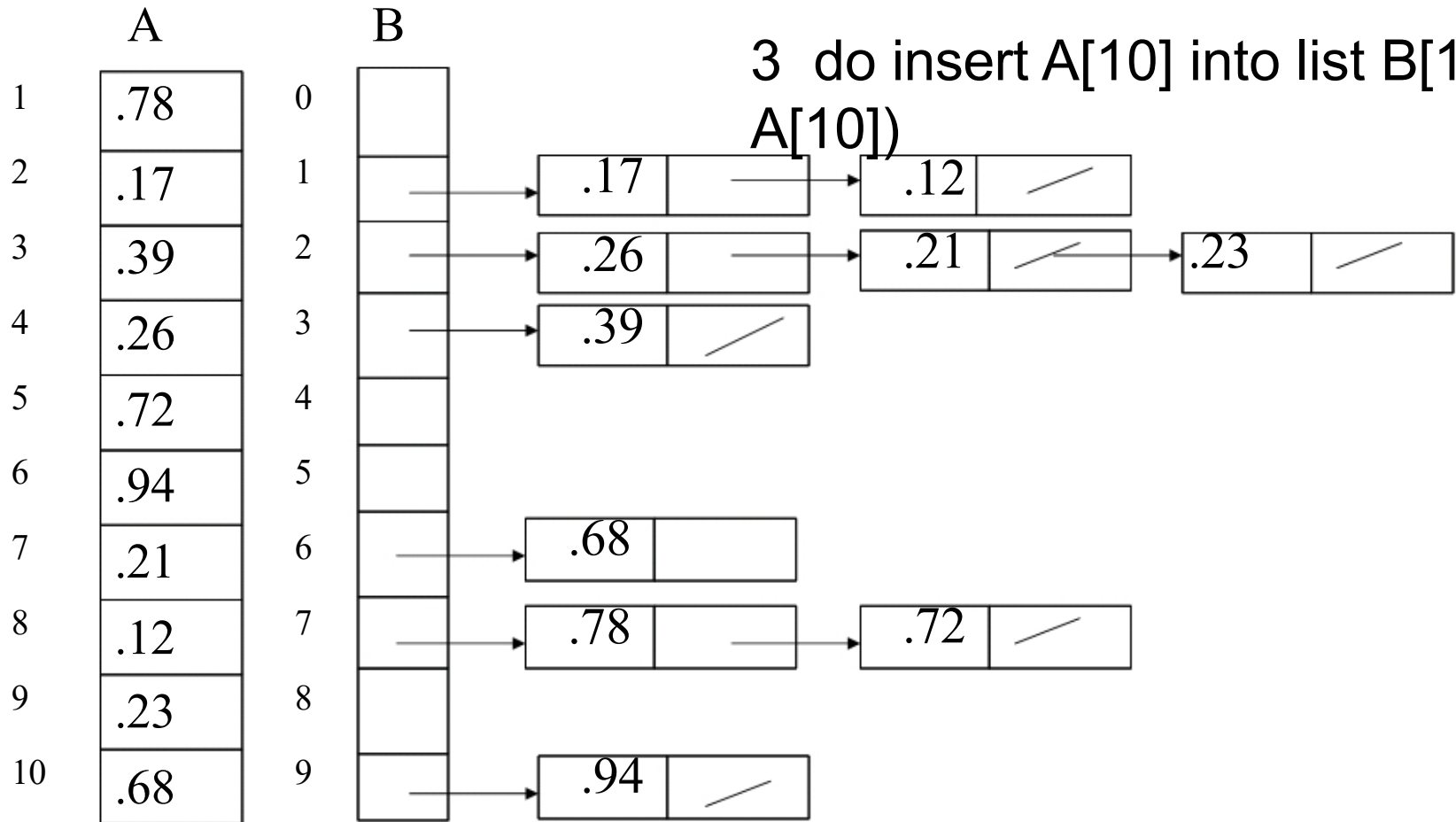
3 do insert $A[9]$ into list $B[10$
 $A[9])$



1 $n = 10$

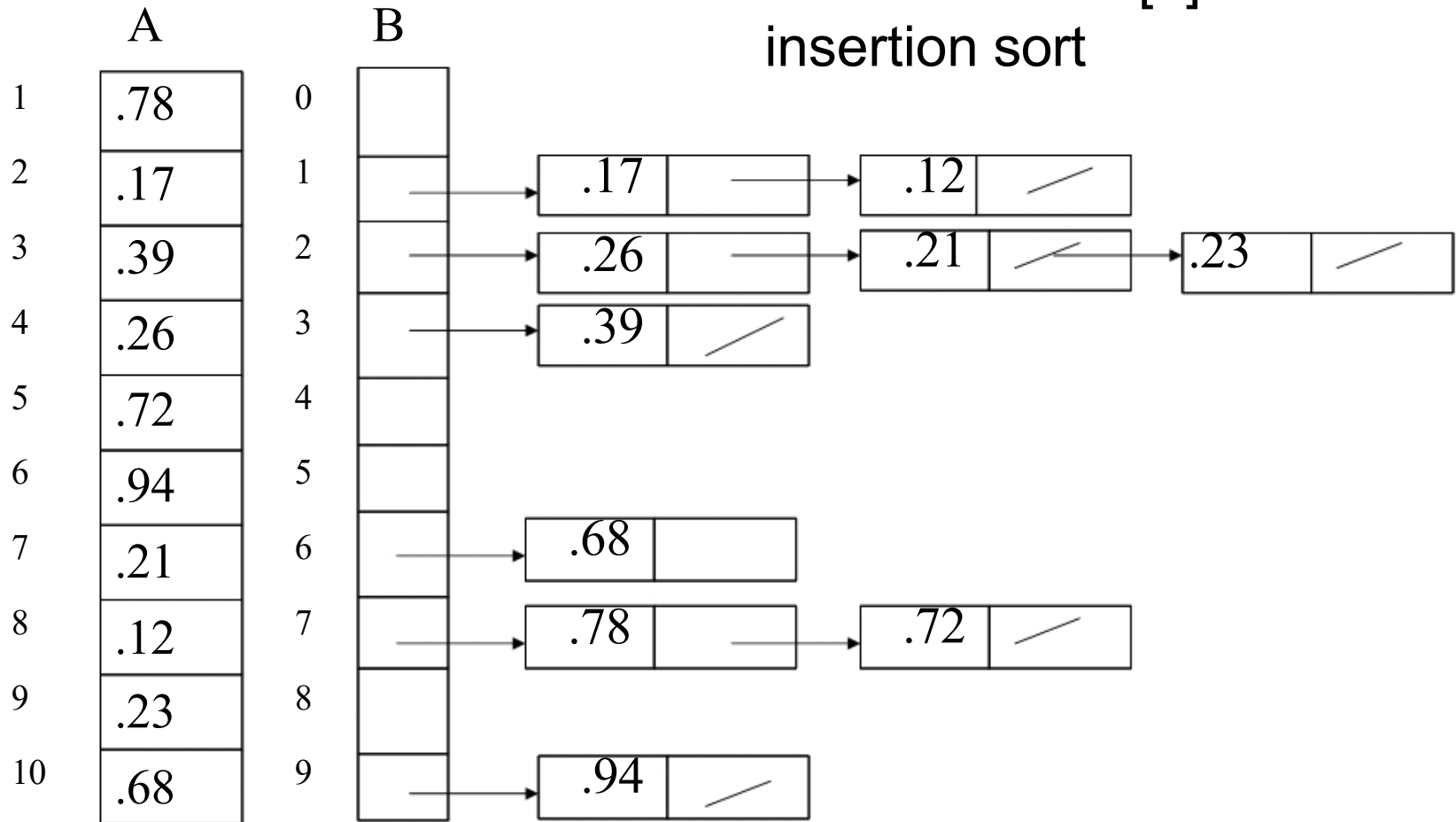
2 for $i = 10$

3 do insert $A[10]$ into list $B[10$
 $A[10])$



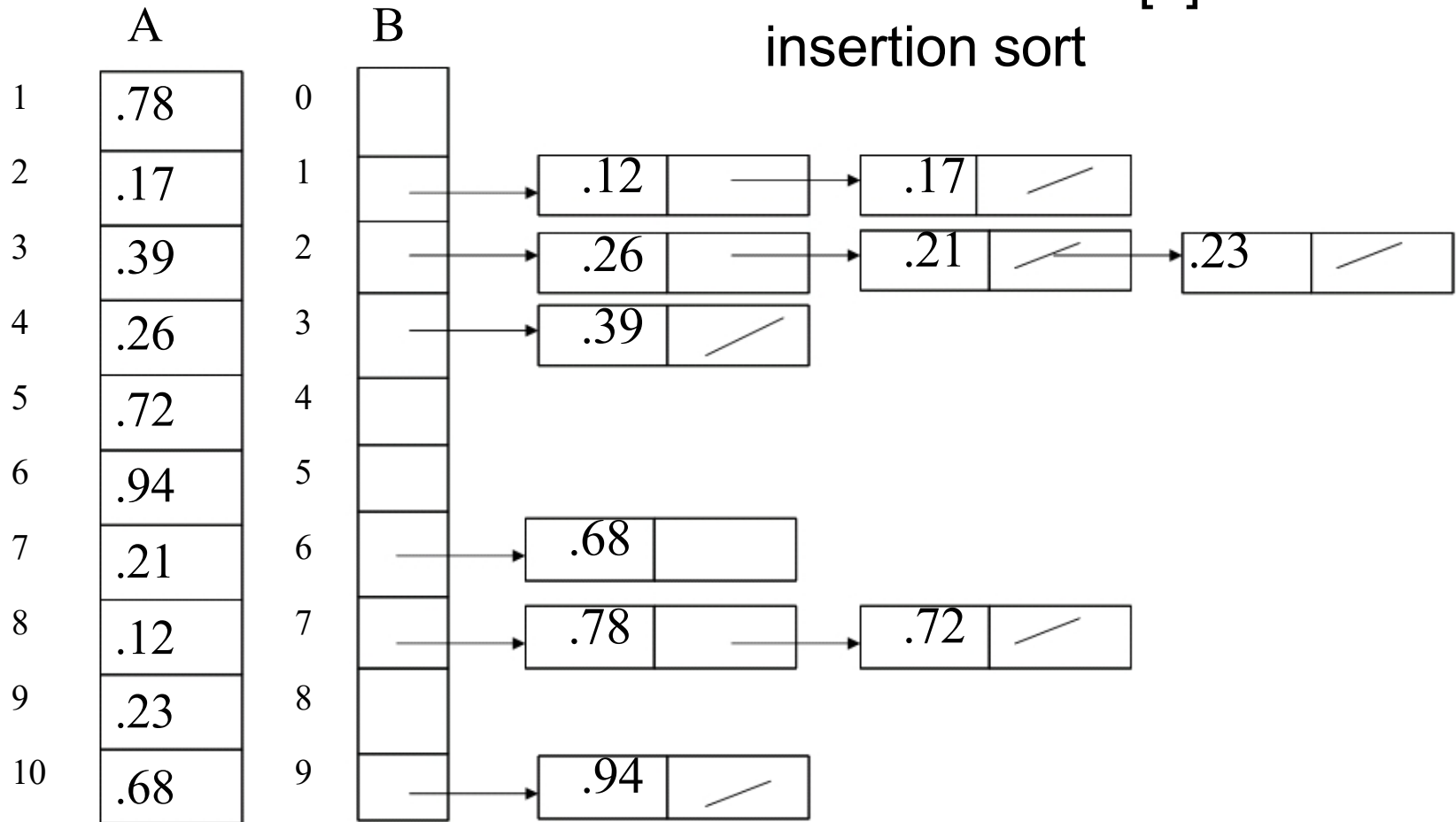
4 $i = 0$

5 do sort list B[0] with
insertion sort



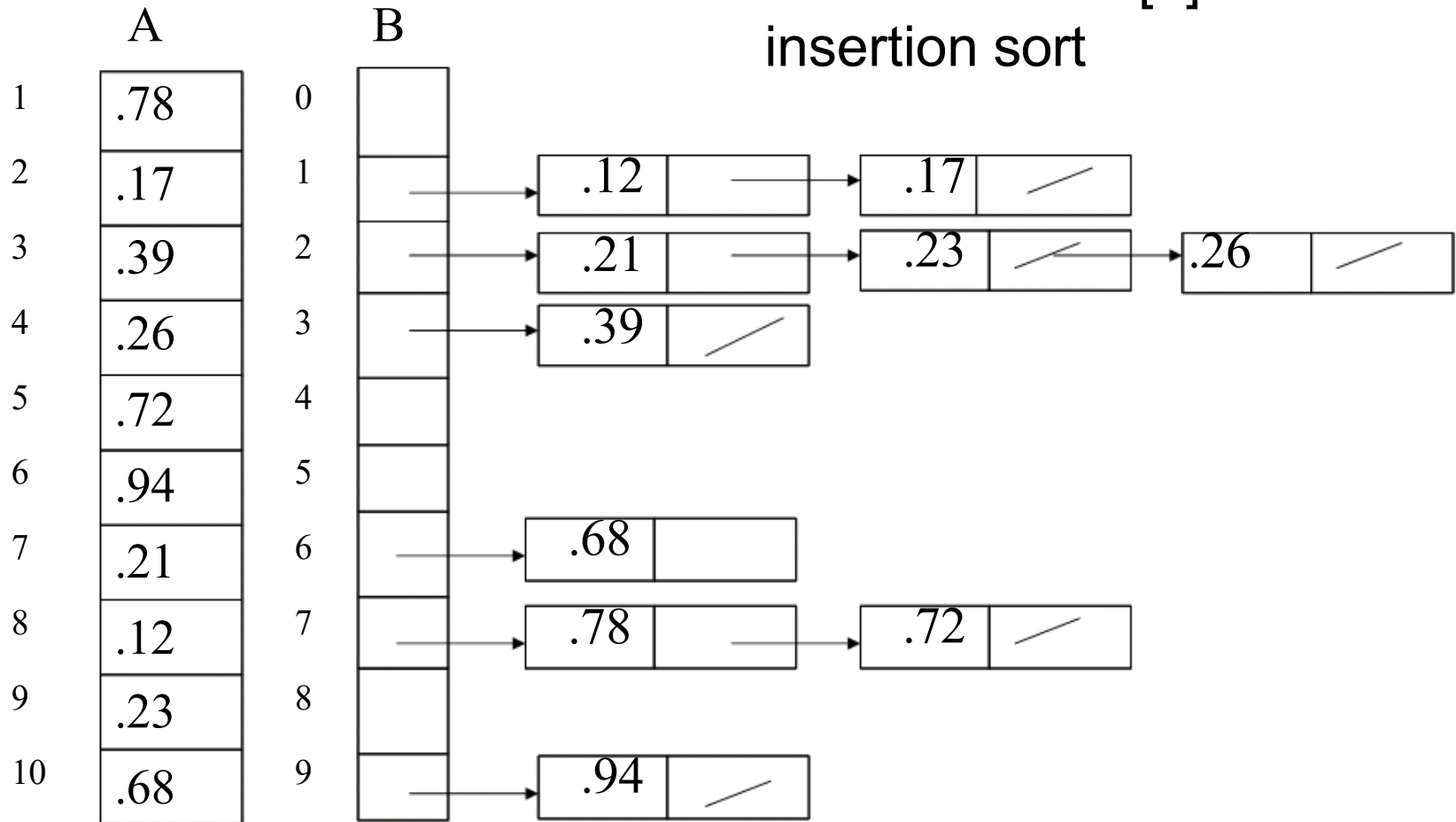
4 $i = 1$

5 do sort list B[1] with
insertion sort



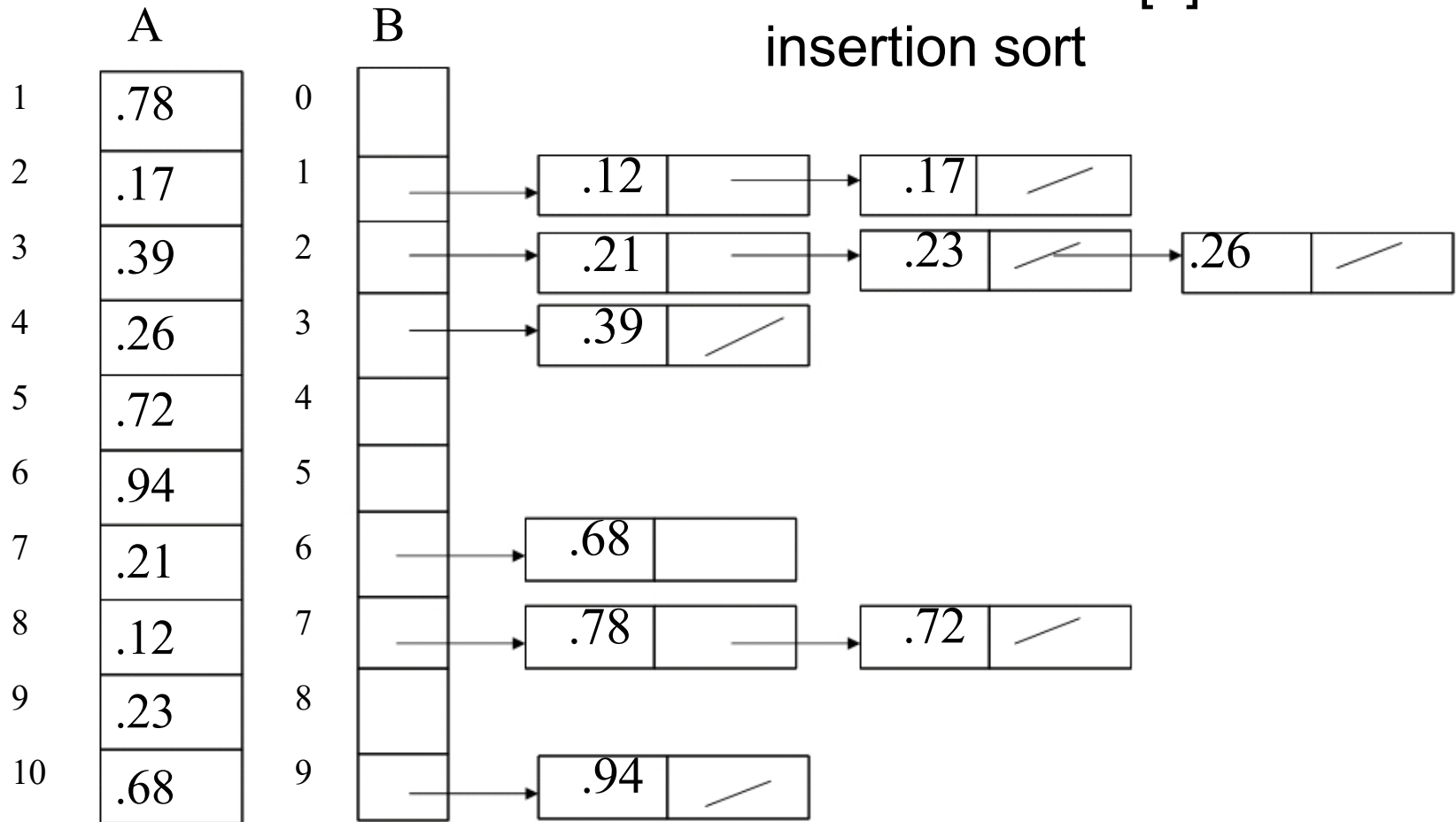
4 $i = 2$

5 do sort list B[2] with
insertion sort



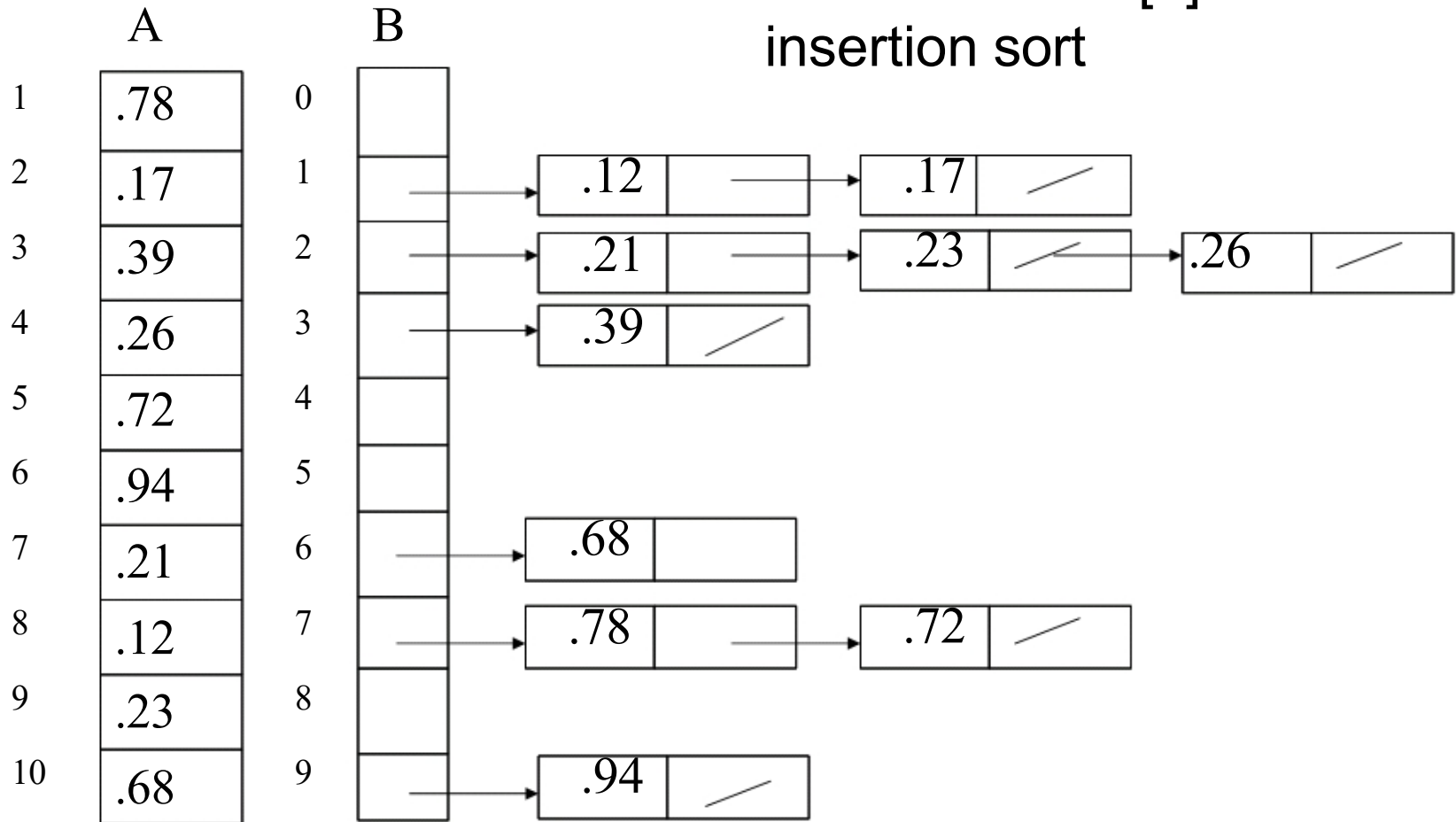
4 $i = 3$

5 do sort list B[3] with
insertion sort



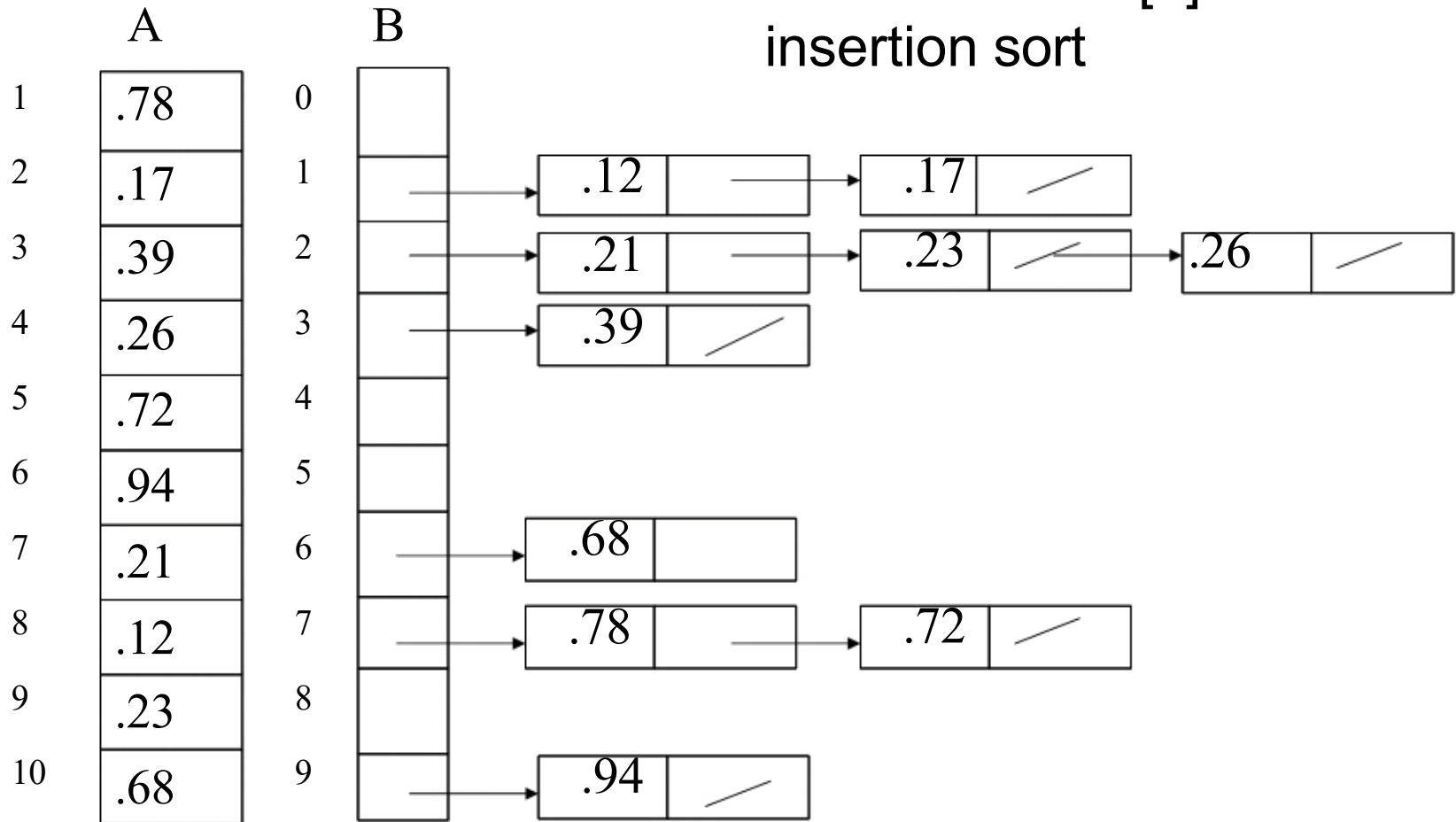
4 $i = 4$

5 do sort list B[4] with
insertion sort



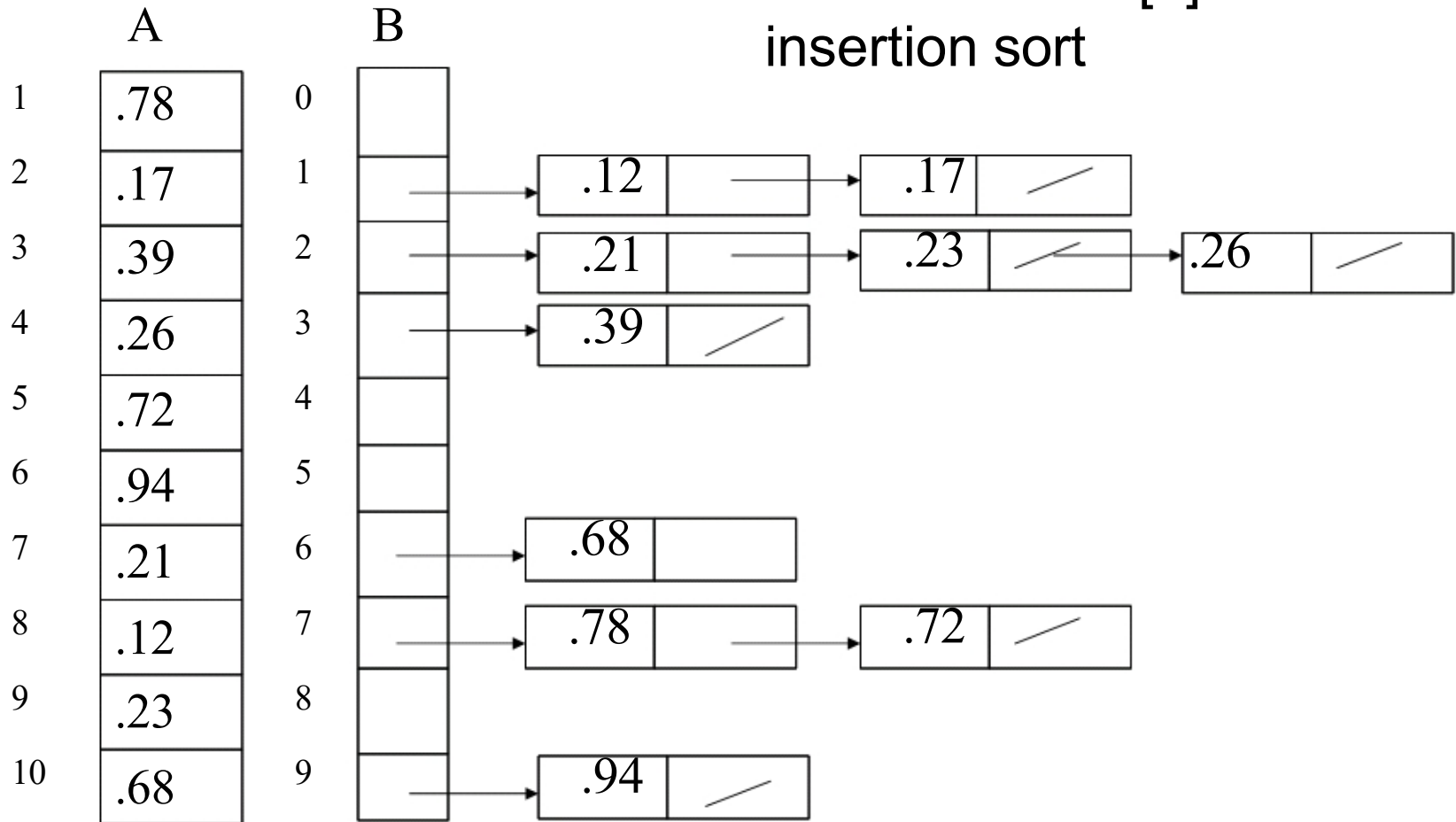
4 $i = 5$

5 do sort list B[5] with
insertion sort



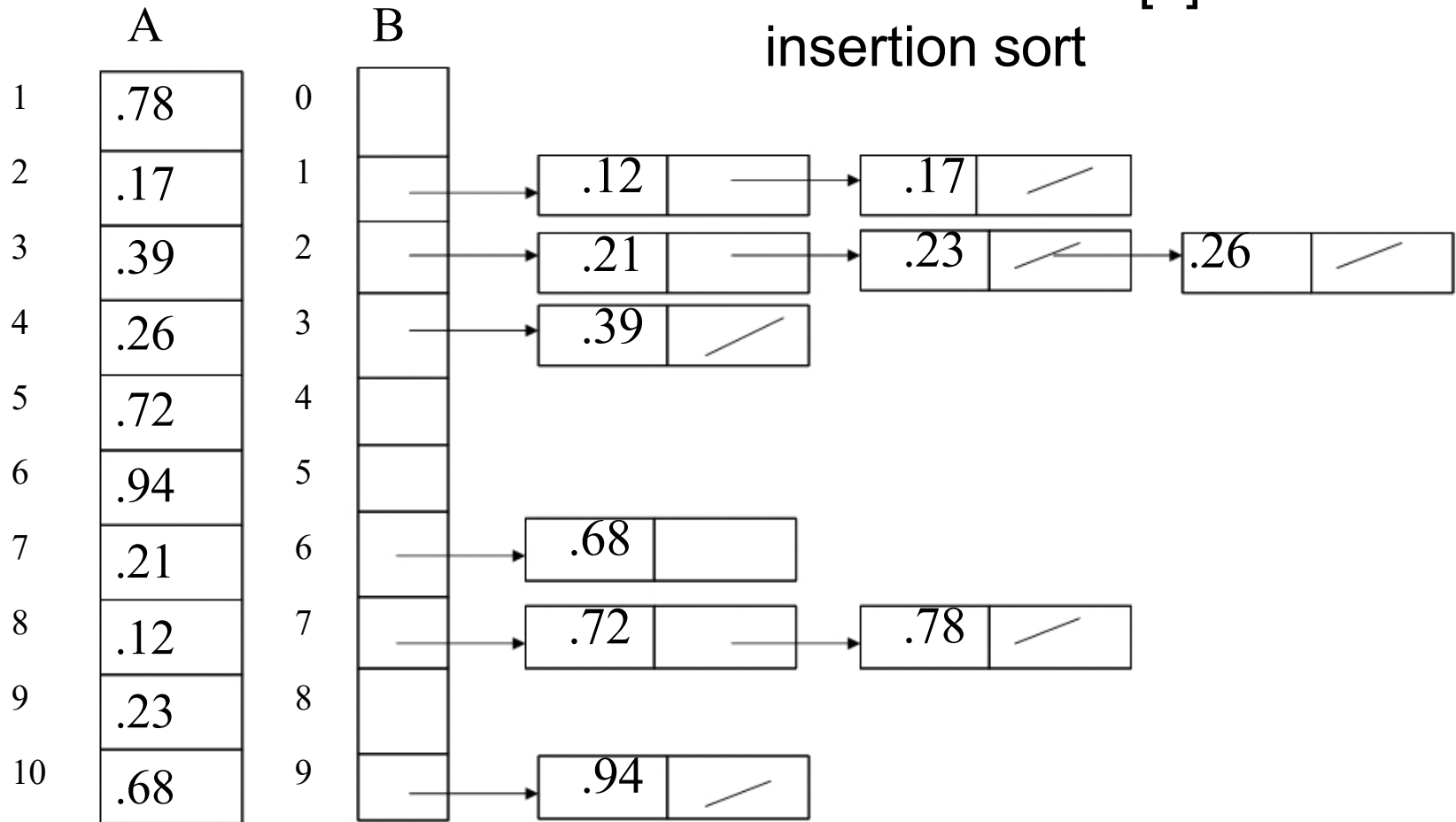
4 $i = 6$

5 do sort list B[6] with
insertion sort



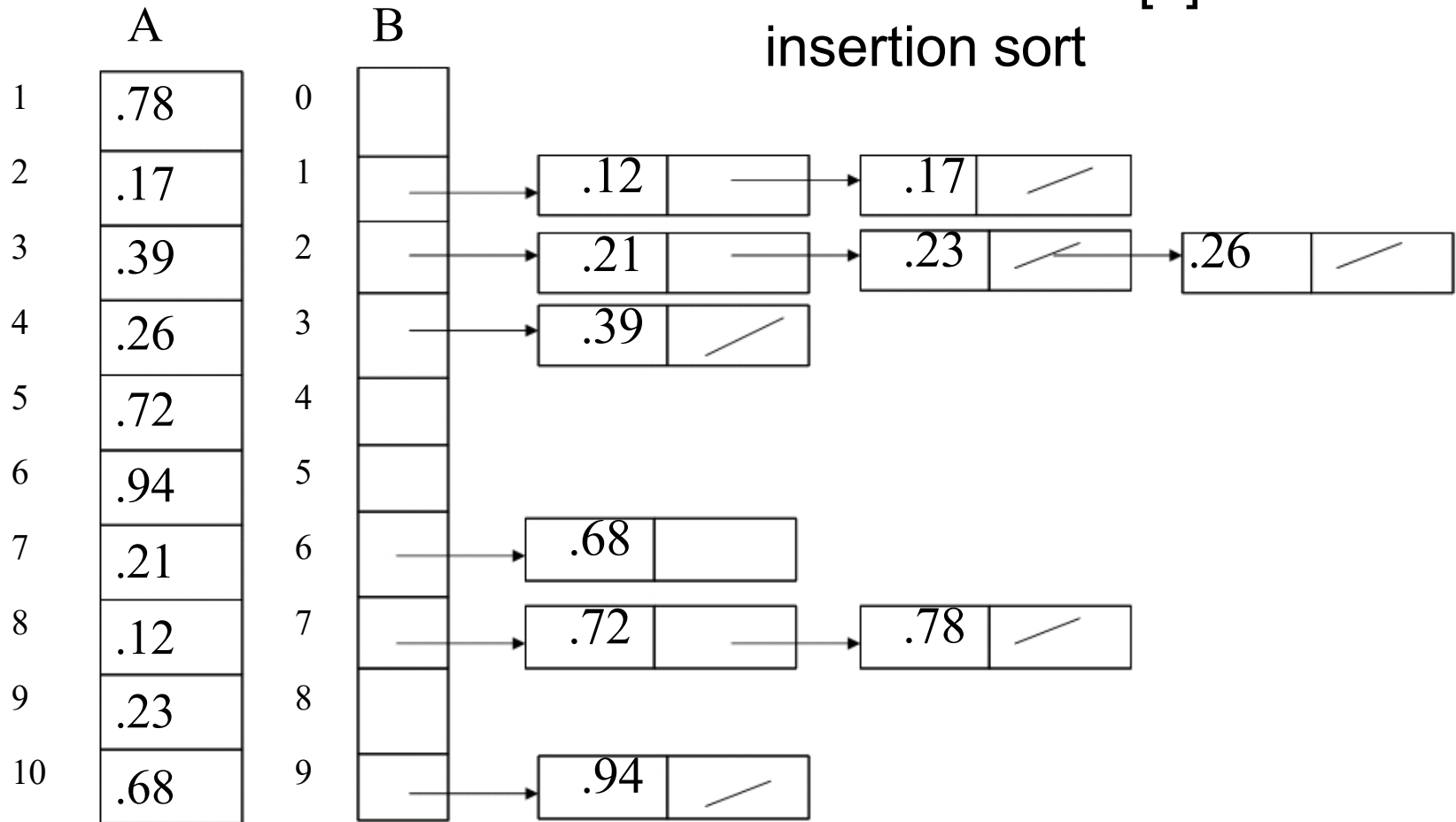
4 $i = 7$

5 do sort list B[7] with
insertion sort



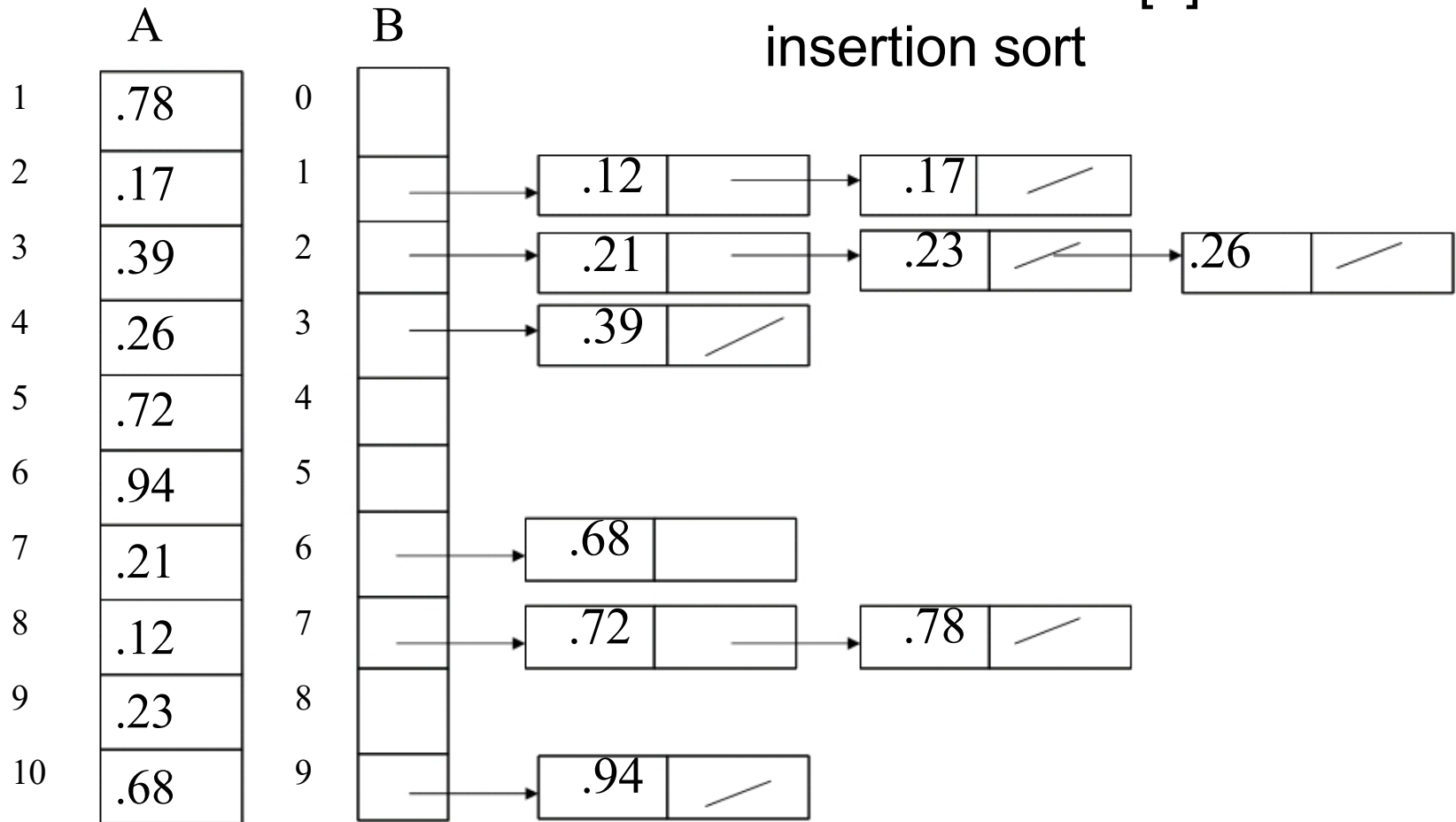
4 $i = 8$

5 do sort list B[8] with
insertion sort



4 $i = 9$

5 do sort list B[9] with
insertion sort





Bucket Sort: Analysis

- All lines except line 5 (insertion sort) takes $O(n)$ time in worst case
- n_i be the random variable denoting number of elements in the bucket $B[i]$
- Expected time to sort the elements in Bucket $B[i]$ is $=O(n_i)^2$



Complexity Analysis of Bucket Sort

The time complexity of bucket sort depends on:

1. The distribution of the input data.
2. The number of buckets used.
3. The sorting algorithm used to sort the individual buckets.

Based on the given factors, bucket sort has the following complexities:

Best Case Time Complexity	$O(n + k)$
Worst Case Time Complexity	$O(n^2)$
Average Case Time Complexity	$O(n + k)$
Space Complexity	$O(n + k)$

Here,

- n is the number of elements.
- k is the number of buckets.